

Mapping circuit to transfer function

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1 Questions Answered

1.1 solving RC circuit : rc1

so we have a RC circuit, we find its transfer function

$$\begin{aligned}H(s) &= \frac{V_{out}(s)}{V_{in}(s)} \\V_{out}(s) &= V_{in}(s) * \frac{Z_c}{Z_c + Z_R} \\V_{out}(s) &= V_{in}(s) * \frac{\frac{1}{sC}}{\frac{1}{sC} + R} \\H(s) &= \frac{\frac{1}{sC}}{\frac{1}{sC} + R}\end{aligned}$$

$V_{in}(s) = 1$ because input is a AC source with amplitude of 1
say

$$C = 1 \text{ pF} = 1 \times 10^{-12} \text{ F}, R = 1 \text{ k}\Omega = 1 \times 10^3 \Omega$$

so now to calculate for the -3dB frequency.
The definition of -3dB frequency is :

The point where a circuit's output power drops to half of its maximum value

in other words :

The point where a circuit's output voltage drops to $\frac{1}{\sqrt{2}}$ of its maximum value

We are looking for the frequency where if we input a cosine wave at such frequency, aka multiply that input expression(say $V_{in}(s)$) with the transfer function $H(s)$, we would get an expression of the $V_{out}(s)$ where

$$V_{out}(s) = V_{in}(s) * |H(\text{frequency of max gain})| * \frac{1}{\sqrt{2}} \quad (1)$$

$$\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2} \quad (2)$$

if $|H(\text{frequency of max gain})| = 1$

@ -3dB frequency :

$$\frac{s}{s^2 + \omega^2} * H(s) = \frac{s}{s^2 + \omega^2} * \frac{1}{\sqrt{2}}$$

looking at (1) we can see that by looking for the -3dB frequency, we're basically looking for where

The point where a circuit's voltage gain drops to $\frac{1}{\sqrt{2}}$ of its maximum voltage gain

now we can actually start calculating the -3dB frequency. Because this is an RC circuit, we know that the max gain frequency is at DC.

$$|H(0)| = \frac{\frac{1}{sC}}{\frac{1}{sC} + R} = \frac{1}{1 + R * sC} = 1$$

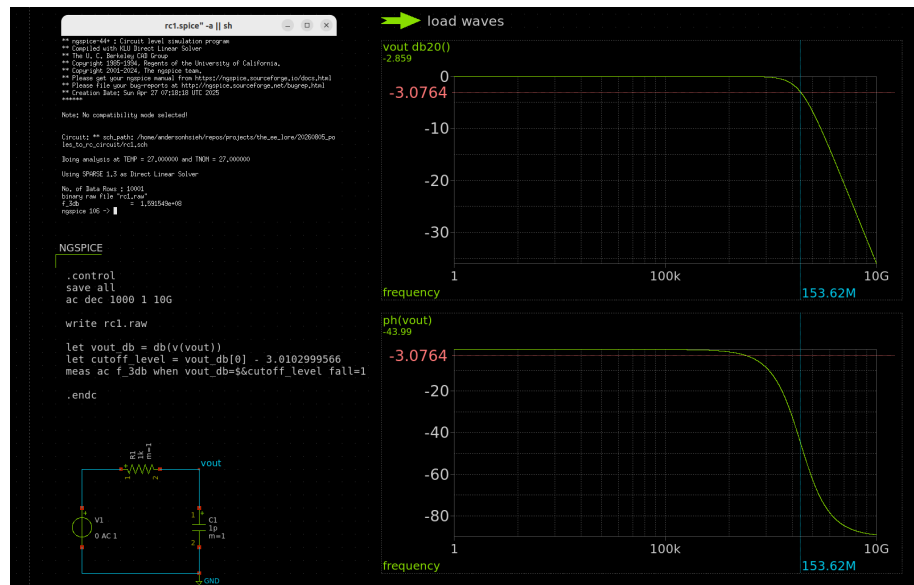
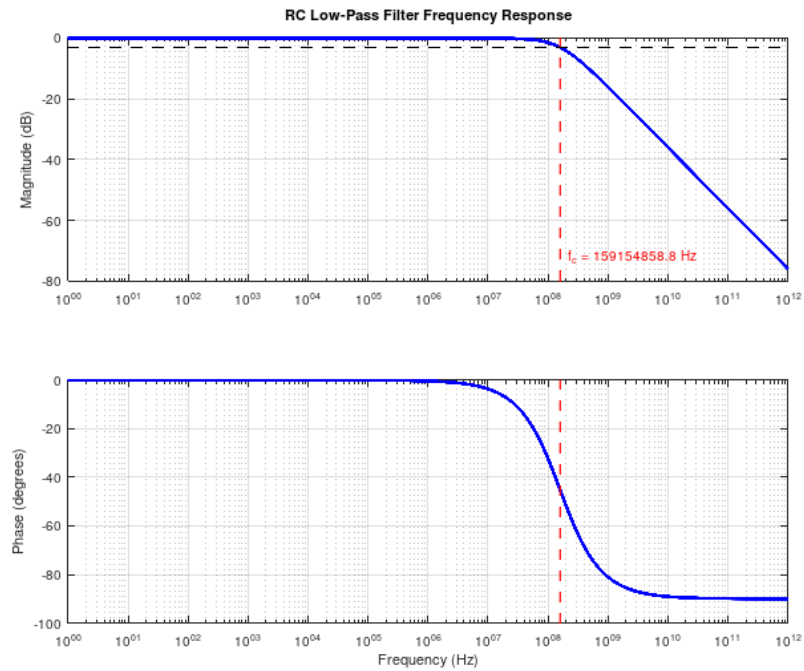
we're looking for the frequency where

$$\begin{aligned}
s &= j\omega \\
|H(s)| &= |H(0)| * \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \approx 0.7071 \\
|H(s)| &= \left| \frac{\frac{1}{sC}}{\frac{1}{sC} + R} \right| \\
\frac{1}{\sqrt{2}} &= \left| \frac{1}{1 + R * sC} \right| \\
\frac{1}{\sqrt{2}} &= \left| \frac{1}{1 + R * j\omega C} \right| \\
\frac{1}{\sqrt{2}} &= \frac{1}{|1 + R * j\omega C|} \\
\frac{1}{\sqrt{2}} &= \frac{1}{\sqrt{1^2 + (\omega RC)^2}} \\
\sqrt{1 + (\omega RC)^2} &= \sqrt{2} \\
1 + (\omega RC)^2 &= 2 \\
(\omega RC)^2 &= 1 \\
\omega RC &= 1 \\
\omega &= 2\pi f = \frac{1}{RC} \\
f_{-3dB} &= \frac{1}{2\pi RC} \\
f_{-3dB} &= \frac{1}{2\pi * 1 \times 10^3 \Omega * 1 \times 10^{-12} \text{ F}} \\
f_{-3dB} &\approx 159.15 \text{ MHz}
\end{aligned} \tag{3}$$

we saw equation (3) is the well known cut off frequency expression for an RC circuit. Plugging our RC numbrers in we get

$$\begin{aligned}
\frac{1}{RC} &= \frac{1}{(1 \times 10^3 \Omega)(1 \times 10^{-12} \text{ F})} = 1 \times 10^9 \text{ rad s}^{-1} \\
f_{-3dB} &= \frac{\omega_{-3dB}}{2\pi} = \frac{1 \times 10^9 \text{ rad s}^{-1}}{2\pi} \approx 159.15 \text{ MHz}
\end{aligned}$$

That reflects our SPICE simulation and Octave script



2 Questions To Be Answered

1. how do they tune VDD for ring oscillator in a real application?

2. poles and zeros in oscillators
3. current mirrors in differential ring oscillator

3 Some Picture

A Derivation of the -3dB Voltage Ratio

For a purely resistive load,

$$P_{\text{load, avg}} = \frac{V_{\text{out, rms}}^2}{R_{\text{load}}}.$$

at -3dB cut off frequency, power delivered to a load with resistance R drops to half.

$$\begin{aligned} \frac{P_{\text{current}}}{P_{\text{max}}} &= \frac{1}{2} \\ \frac{P_{\text{current}}}{P_{\text{max}}} &= \frac{1}{2} \\ \left(\frac{V_{\text{current}}}{V_{\text{max}}} \right)^2 &= \frac{1}{2} \\ \frac{V_{\text{current}}}{V_{\text{max}}} &= \sqrt{\frac{1}{2}} \\ &= \frac{1}{\sqrt{2}} \\ &\approx 0.7071 \end{aligned}$$

B Why Use V_{rms} When Calculating Average Power?

RMS voltage is used because it represents the DC voltage that would produce the same average power, or heating, in a resistor.

The instantaneous power dissipated by a resistor is

$$p(t) = \frac{v(t)^2}{R}.$$

For a sinusoidal voltage,

$$v(t) = V_{\text{peak}} \cos(\omega t),$$

so the instantaneous power is

$$p(t) = \frac{V_{\text{peak}}^2 \cos^2(\omega t)}{R}.$$

Since the average value of $\cos^2(\omega t)$ over one complete cycle is $1/2$, the average power is

$$P_{\text{avg}} = \frac{V_{\text{peak}}^2}{2R}.$$

But if whenever we want to calculate average power we gotta divide by 2 it's too much work, so to make it easier when calculating average power over a complete cycle of a sinusoidal signal, we define a new kind of voltage, the RMS voltage

The RMS voltage of a sine wave is

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}}.$$

Therefore, when we want to find average power of a sinusoidal signal we can just divide the peak voltage by $\sqrt{2}$ and plug that into the power = $I \cdot V$ equation

$$P_{\text{avg}} = \frac{V_{\text{rms}}^2}{R}.$$

For example, a sine wave with a peak voltage of 1 V has an RMS voltage of approximately 0.7071 V. Across a 1Ω resistor, its average power is

$$\begin{aligned} P_{\text{avg}} &= \frac{(0.7071 \text{ V})^2}{1 \Omega} \\ &\approx 0.5 \text{ W}. \end{aligned}$$

Thus, an AC voltage of 0.7071 V RMS produces the same average heating in a resistor as if there is a DC voltage of 0.7071 V across the resistor.